

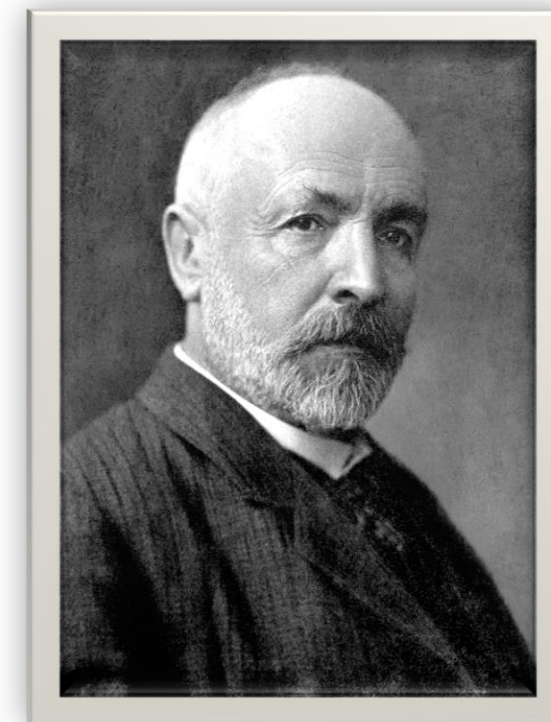
Math Primer

(1. Set theory basics)

Contents



- ▶ Sets and their Representations
- ▶ The Empty Set
- ▶ Finite & Infinite Sets
- ▶ Equal Sets
- ▶ Subsets
- ▶ Cardinality and Power Set
- ▶ Operations on Set



Georg Cantor
(1845-1918)



Sets and their Representations

- ▶ Set is a basic 'undefined' fundamental part of mathematics
- ▶ Understood as a collection of well-defined elements (e.g. odd numbers, prime numbers < 100 ...)
- ▶ In Maths, the following sets are referenced all the time

$\mathbb{N}$: the set of all natural numbers

$\mathbb{Z}$: the set of all integers

$\mathbb{Q}$: the set of all rational numbers

$\mathbb{R}$: the set of all real numbers

$\mathbb{C}$: the set of complex numbers

- ▶ Set of all positive real numbers : $\mathbb{R}^+$

Sets and their Representations

- ▶ A set can be represented using
 - ▶ Tabular form (a comma separated list of all elements)
 - ▶ Set builder form (based on a common rule)
- ▶ For example : the set of factors of 42 can be written as
 - Tabular form : $A = \{1, 2, 3, 6, 7, 14, 21, 42\}$
 - Set builder form : $A = \{x : x \text{ is a factor of } 42\}$
- ▶ Set A contains elements 1, 2, 3, 6, 7, 14, 21, 42
- ▶ 1 belongs to / a member of set A. It's written as $1 \in A$
- ▶ 4 does not belong to set A. It's written as $4 \notin A$
- ▶ Elements are not repeated (no duplicates)
- ▶ Ordering of elements is immaterial



Sets and their Representations

Exercise 1 : Write the solution set of the equation $x^2 + x - 2 = 0$ in tabular form

Exercise 2 : Write the set $\{x : x \text{ is a positive integer and } x^2 < 40 \}$ in the tabular form

Exercise 3 Write the set $A = \{1, 4, 9, 16, 25, \dots\}$ in set-builder form

Exercise 4 Match the following

Tabular form

(i) $\{P, R, I, N, C, A, L\}$

(ii) $\{0\}$

(iii) $\{1, 2, 3, 6, 9, 18\}$

(iv) $\{3, -3\}$

Set builder form

(a) $\{x : x \text{ is a positive integer and is a divisor of } 18\}$

(b) $\{x : x \text{ is an integer and } x^2 - 9 = 0\}$

(c) $\{x : x \text{ is an integer and } x + 1 = 1\}$

(d) $\{x : x \text{ is a letter of the word PRINCIPAL}\}$



The Empty Set

- ▶ A set that has no elements is called *empty set* or *null set* or *void set*
- ▶ e.g., $A = \{ x : x \text{ is even and odd} \}$, $B = \{ x : x \in \mathbb{N} \text{ and } 1 < x < 2 \}$
- ▶ It's denoted as ϕ or $\{ \}$



Finite and Infinite Sets

- ▶ Finite Sets have fixed number of elements
- ▶ Infinite Sets have infinite number of elements

Exercise 5 : Classify the following as finite vs infinite

- (i) set of the days of the week.
- (ii) set of solutions of the equation $x^2 - 16 = 0$.
- (iii) set of points on a line.
- (iv) $\mathbb{N}$

- ▶ It's possible to write all elements of Finite Sets in roster form
- ▶ When writing Infinite sets in roster form, we use three dots e.g. set of odd numbers $\{ 1, 3, 5, \dots \}$



Equal Sets

- ▶ Sets A and B are equal if every element of A is in B and vice versa.
- ▶ We write $A = B$

Exercise 6 : Find pair of sets from the following which are equal

$$A = \{0\},$$

$$B = \{x : x > 15 \text{ and } x < 5\},$$

$$C = \{x : x - 5 = 0\},$$

$$D = \{x : x^2 = 25\},$$

$$E = \{x : x \text{ is the positive root of the equation } x^2 - 2x - 15 = 0\}$$

$$F = \{1, 2, 3\}$$

$$G = \{2, 1, 1, 3, 3\}$$



Subsets and Supersets

- ▶ If every element of set A belongs to set B , then we say A is a subset of B . We write $A \subset B$
- ▶ In other words, $A \subset B$ if $x \in A \Rightarrow x \in B$ (Note $\Rightarrow$ is read as “implies”)
- ▶ If $A \subset B$ and $B \subset A \Leftrightarrow A = B$ (Note $\Leftrightarrow$ means “implies and implied by” or “if and only if” or “iff”)
- ▶ The empty set is the subset of all sets $\phi \subset A$
- ▶ If $A \subset B$ and $A \neq B$, then A is a proper subset of B and B is a superset of A (e.g., $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$, then $A \subset B$, $B \supset A$)
- ▶ Every set is a subset and superset of itself
- ▶ If A has only one element, it's called a singleton. (e.g., $\{1\}$)

Exercise 7

Consider the sets ϕ , $A = \{ 1, 3 \}$, $B = \{1, 5, 9\}$, $C = \{1, 3, 5, 7, 9\}$.

Insert the symbol $\subset$ or $\not\subset$ between each of the following pair of sets:

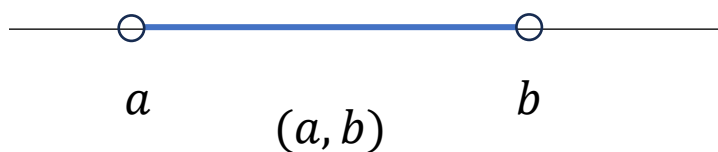
(i) $\phi \dots B$ (ii) $A \dots B$ (iii) $A \dots C$ (iv) $B \dots C$

Set within a Set ?



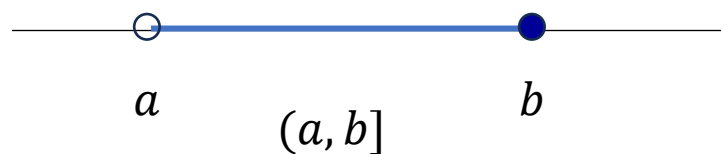
▶ Let A , B and C be three sets. If $A \in B$ and $B \subset C$, is it true that $A \subset C$? If not, give an example.

Intervals as subsets of $\mathbb{R}$



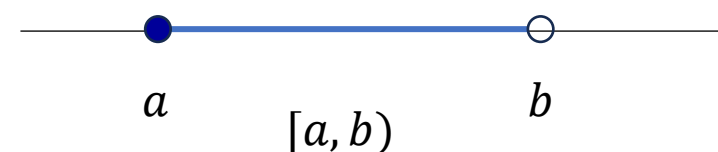
(Open interval)

$$\{x : a < x < b\}$$



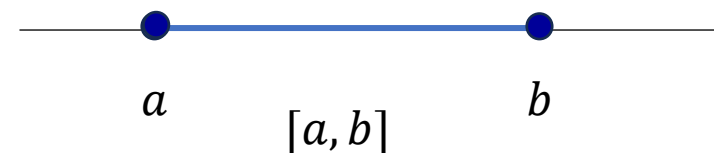
(semi-open interval)

$$\{x : a < x \leq b\}$$



(semi-open interval)

$$\{x : a \leq x < b\}$$



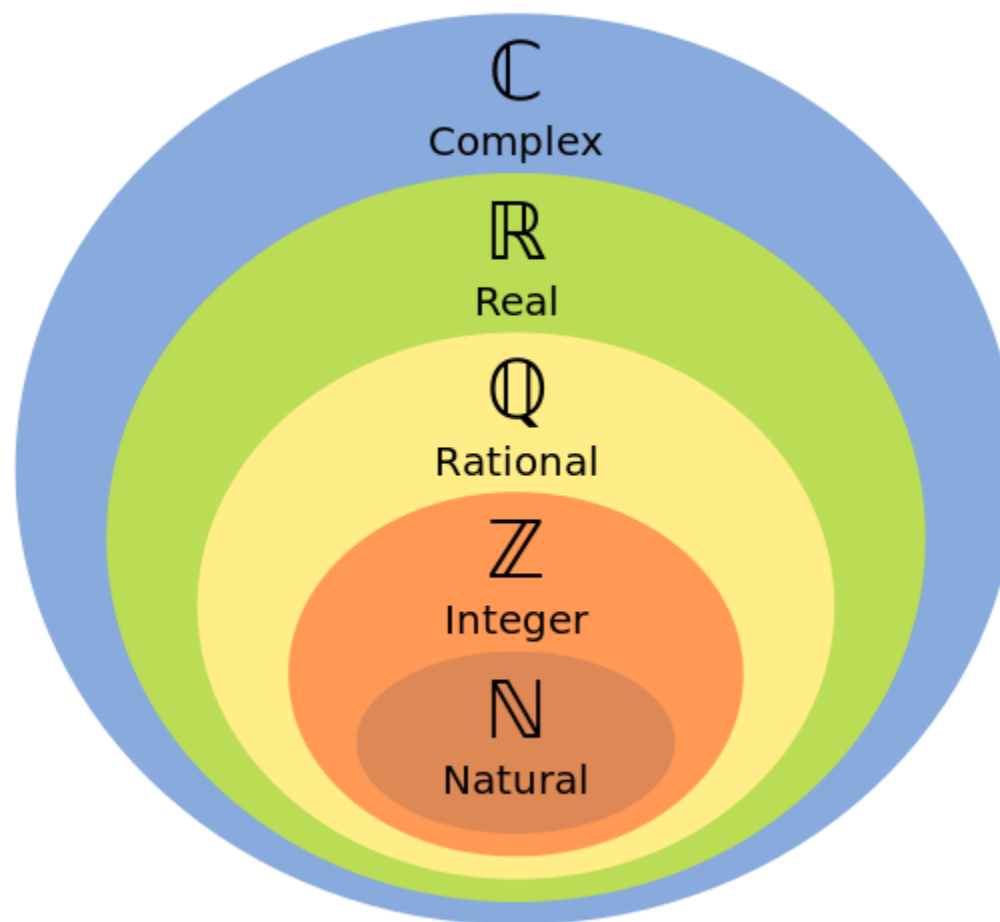
(Closed interval)

$$\{x : a \leq x \leq b\}$$

Subsets of Numbers

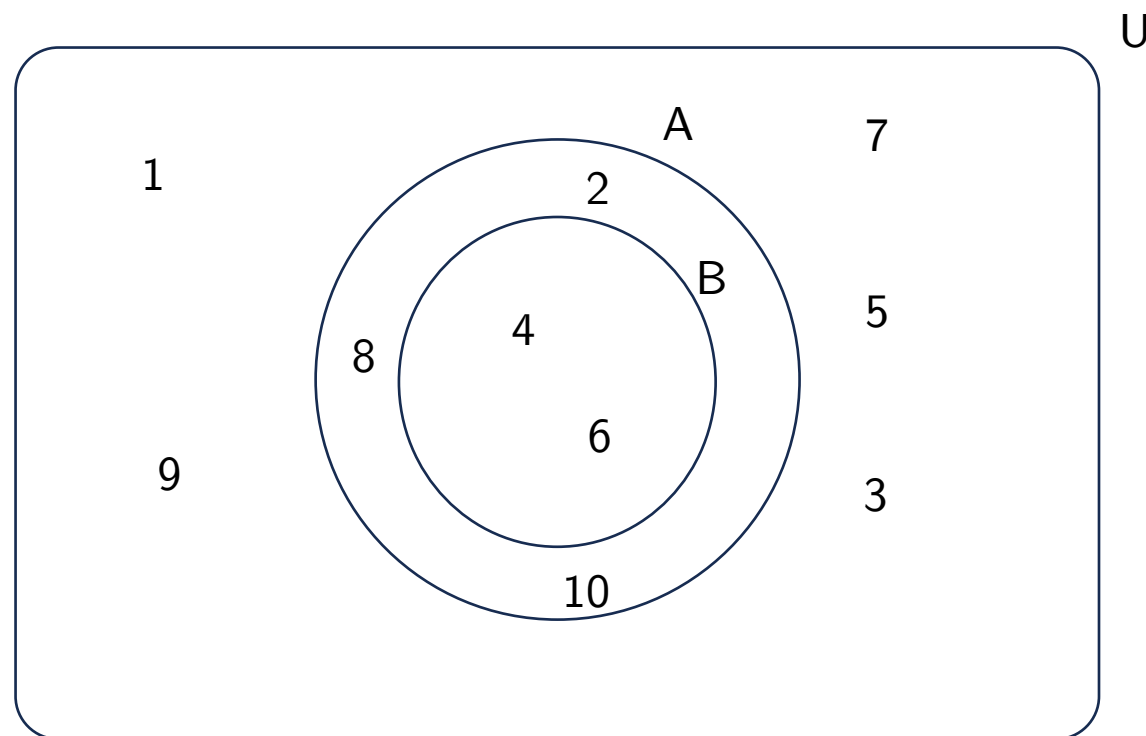


$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$



Venn Diagrams

- ▶ In the example below, $U = \{ 1, 2, 3, \dots, 10 \}$ is the Universal Set
- ▶ A and B are subsets of U i.e. $A \subset U$, $B \subset U$, also $B \subset A$





Cardinality and Power Set

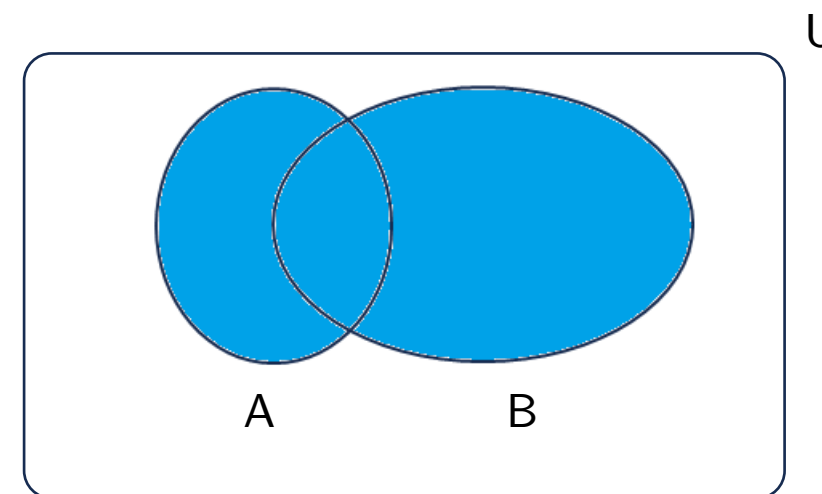
- ▶ Cardinality of a Set is the measure of number of elements in a set
- ▶ e.g., if $A = \{ 2, 4, 6, 8 \}$, $|A| = 4$
- ▶ Power set of a set A is the collection of all sub-sets of A including the empty set and itself
- ▶ e.g., $A = \{ 1, 2 \}$, $\mathcal{P}(A) = \{ \phi, \{1\}, \{2\}, \{1,2\} \}$
- ▶ If $|A| = n$, $|\mathcal{P}(A)| = 2^n$

Exercise 8 Write the power set of $A = \{ 1, 2, 3 \}$

Exercise 9 Write the power set of ϕ . What is $|\mathcal{P}(\phi)|$

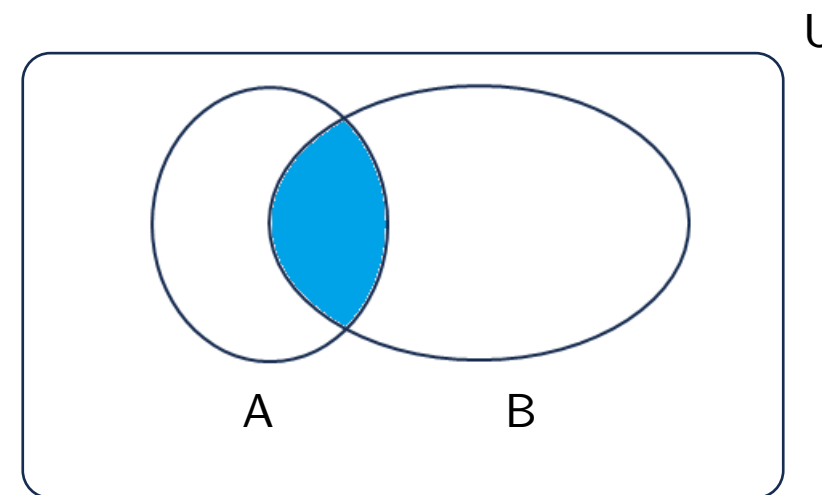
Set Operations - Union

- ▶ Union of sets : $A \cup B$, set of all elements of A and B combined
- ▶ In other words, $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- ▶ e.g., $A = \{2, 4, 6, 8\}$ and $B = \{6, 8, 10, 12\}$, $A \cup B = \{2, 4, 6, 8, 10, 12\}$
- ▶ Some Properties
 - ▶ if $A \cup B = A$, then $B \subset A$
 - ▶ $A \cup \phi = A$
 - ▶ $A \cup A = A$
 - ▶ $A \cup U = U$



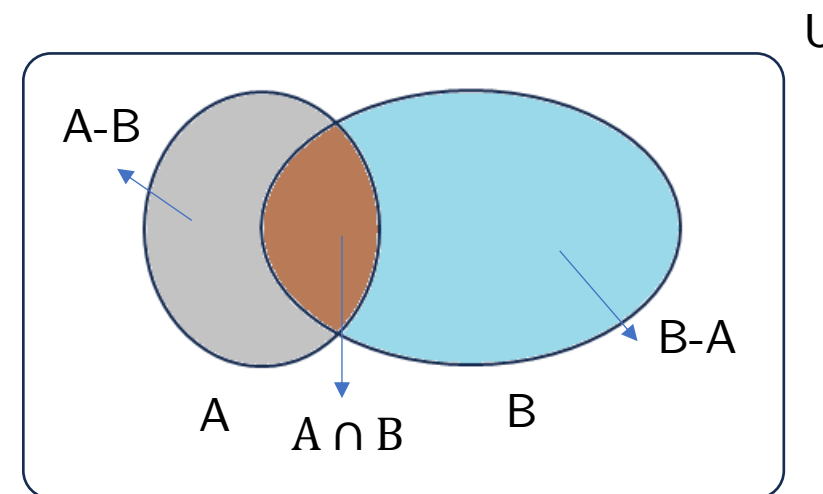
Set Operations - Intersection

- ▶ Intersection of sets : $A \cap B$, set of all elements common to A and B
- ▶ In other words, $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- ▶ e.g., $A = \{2, 4, 6, 8\}$ and $B = \{6, 8, 10, 12\}$, $A \cap B = \{6, 8\}$
- ▶ Some Properties
 - ▶ if $A \cap B = B$, then $B \subset A$
 - ▶ $A \cap \phi = \phi$
 - ▶ $A \cap A = A$
 - ▶ $A \cap U = A$
 - ▶ if $A \cap B = \phi$, A,B are called *dis-joint*
 - ▶ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, distributive



Set Operations - Difference

- ▶ Difference of sets : $A - B$, set of all elements in A but not in B (also written as $A \setminus B$)
- ▶ In other words, $A - B = \{x : x \in A \text{ and } x \notin B\}$
- ▶ e.g., $A = \{2, 4, 6, 8\}$ and $B = \{6, 8, 10, 12\}$, $A - B = \{2, 4\}$, $B - A = \{10, 12\}$
- ▶ Some Properties
 - ▶ $A - B \neq B - A$
 - ▶ $A - \phi = A$
 - ▶ $A - A = \phi$
 - ▶ $A - U = \phi$
 - ▶ $A \cap B, A - B$ and $B - A$ are dis-joint



Set Operations - Complement

- ▶ Complement of a set: A^C , set of all elements in U but not in A . This is same as $A^C = U - A$
- ▶ In other words, $A^C = \{x : x \in U \text{ and } x \notin A\}$
- ▶ e.g., if $A = \{2, 4, 6, 8\}$ and $U = \{1, 2, 3, \dots, 10\}$, $A^C = \{1, 3, 5, 7, 9, 10\}$
- ▶ Some Properties
 - ▶ $(A^C)^C = A$
 - ▶ $A \cup A^C = U$
 - ▶ $A \cap A^C = \phi$
 - ▶ $U^C = \phi$
 - ▶ $\phi^C = U$
- ▶ De Morgan's Law
 - ▶ $(A \cup B)^C = A^C \cap B^C$
 - ▶ $(A \cap B)^C = A^C \cup B^C$

